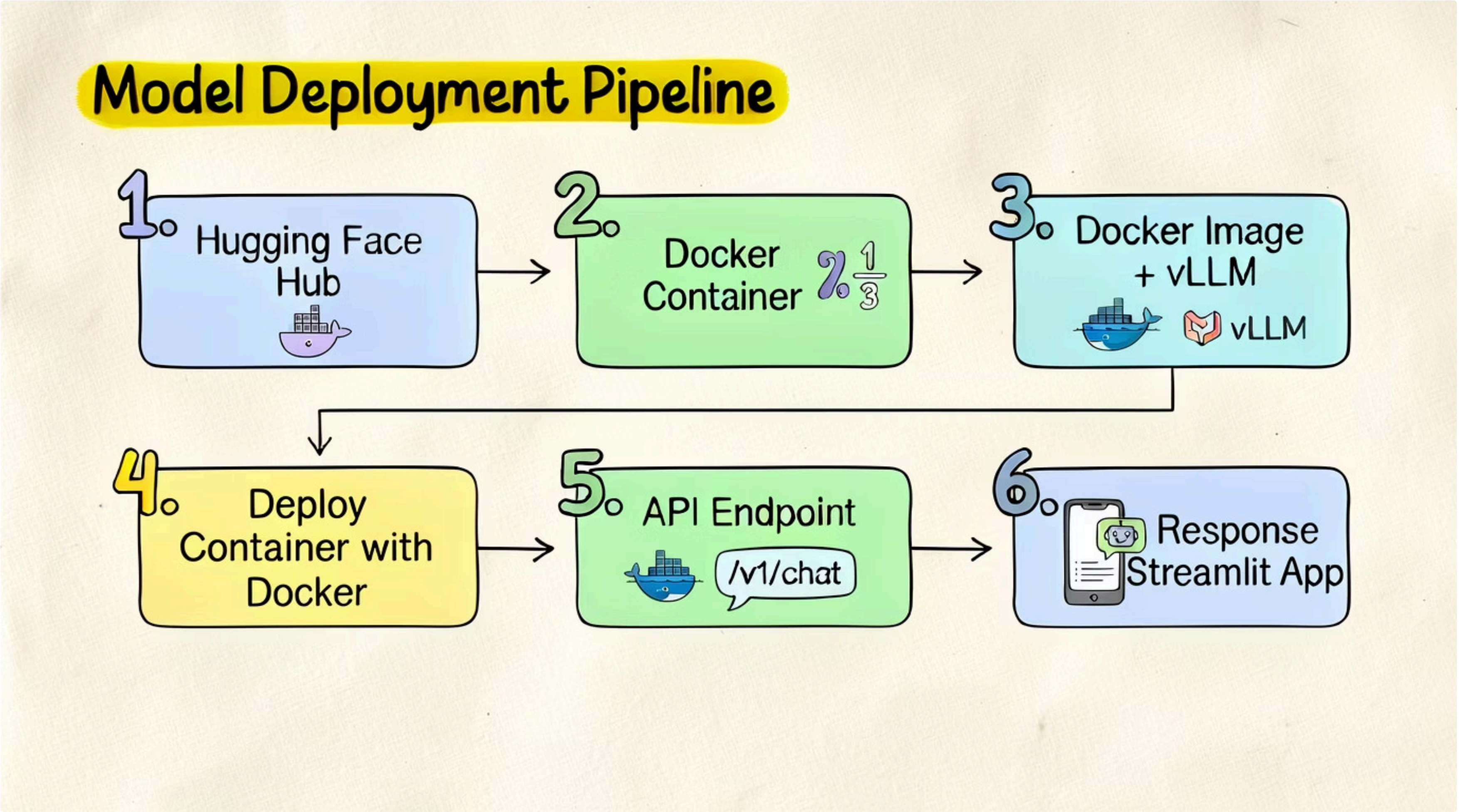
Deploy an LLM on Docker with vLLM

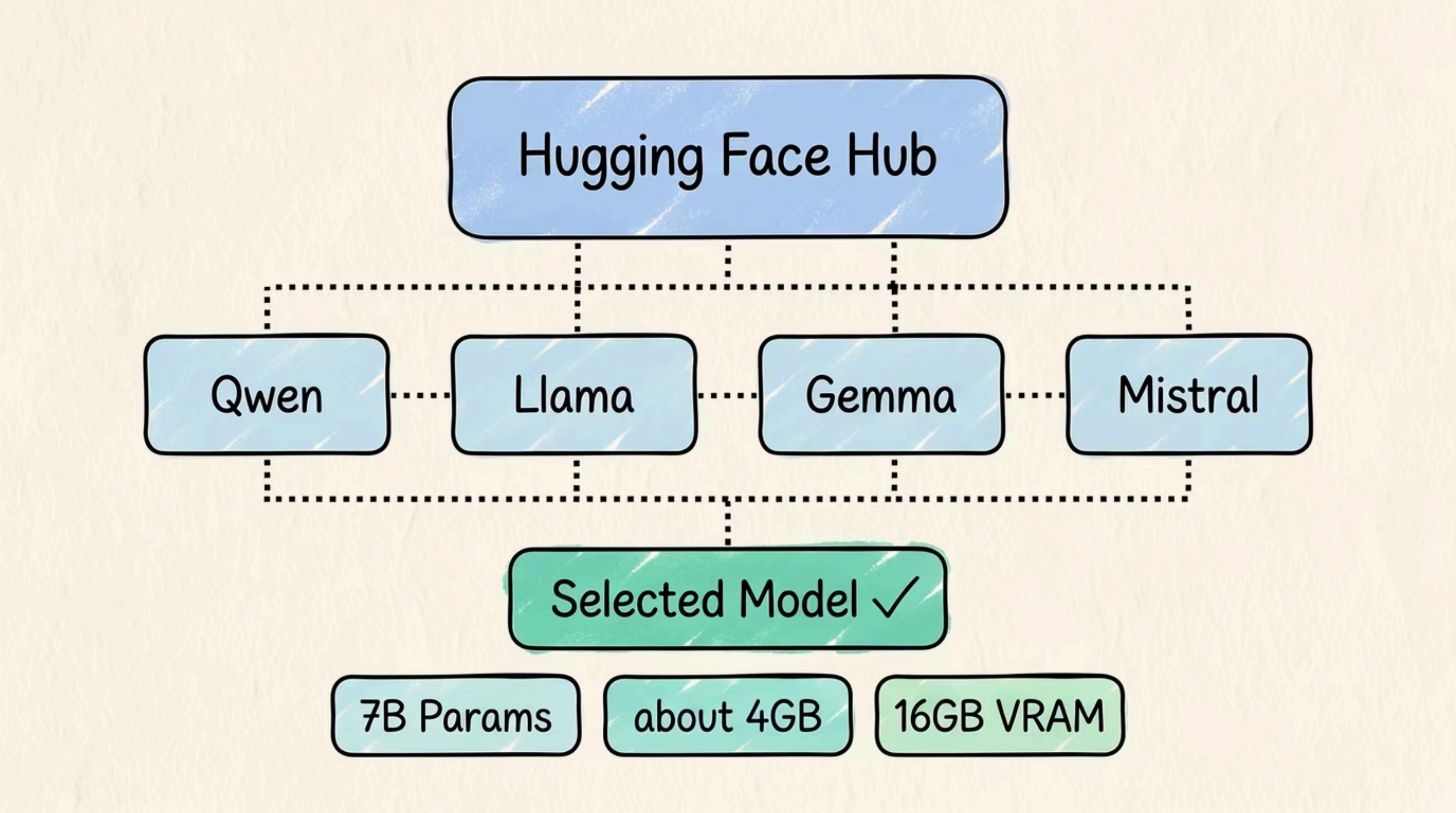


🤗 Step 1

Select the Model

Pick the right open-source LLM before anything else.

> Start small. A 7B model on one GPU beats a 70B model you can't run.

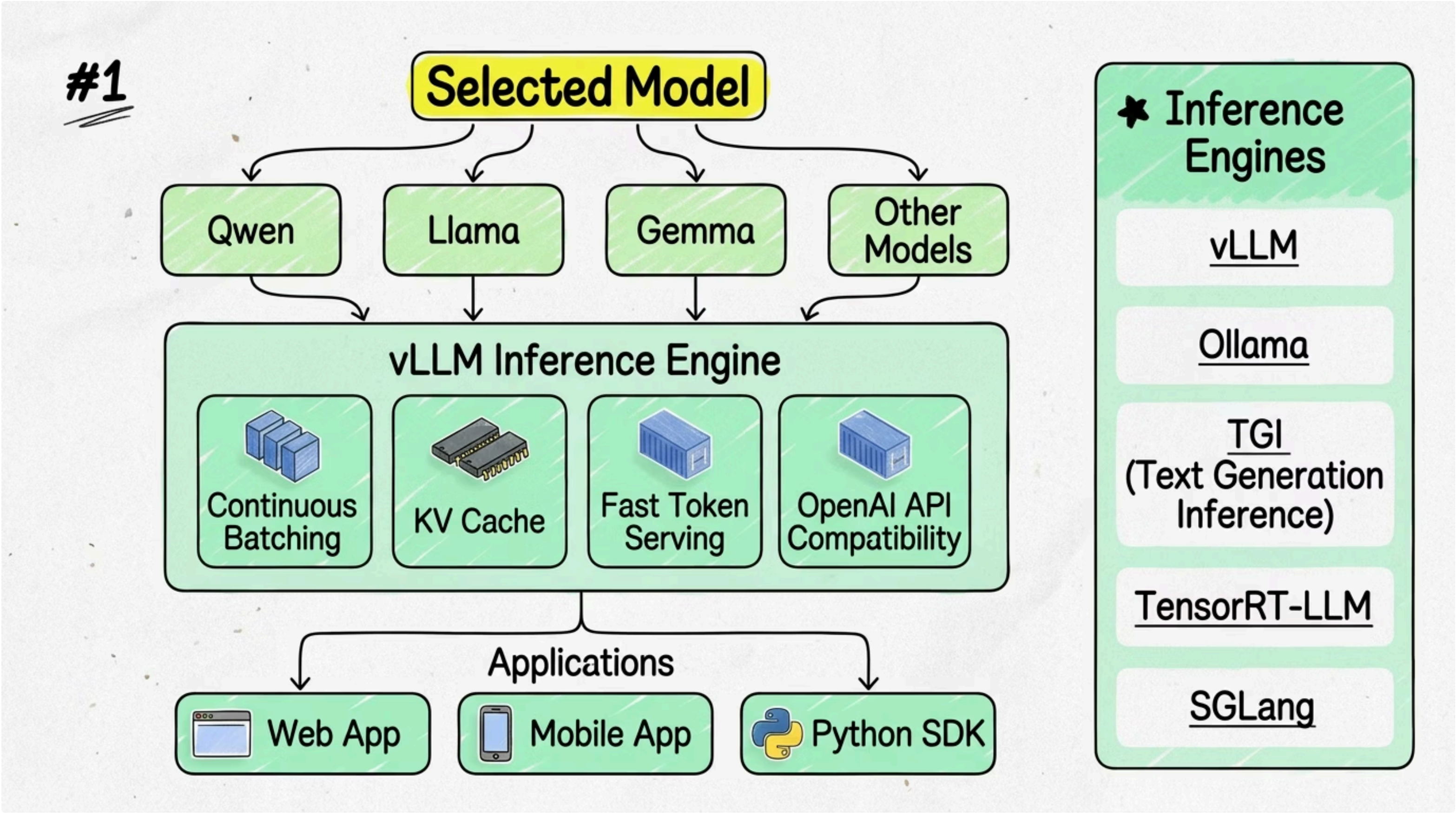


⚙️ Step 2

Choose an Inference Engine

Models need a serving layer before they can handle requests.

> vLLM sits between your model and your users — handling batching, caching, and the API.

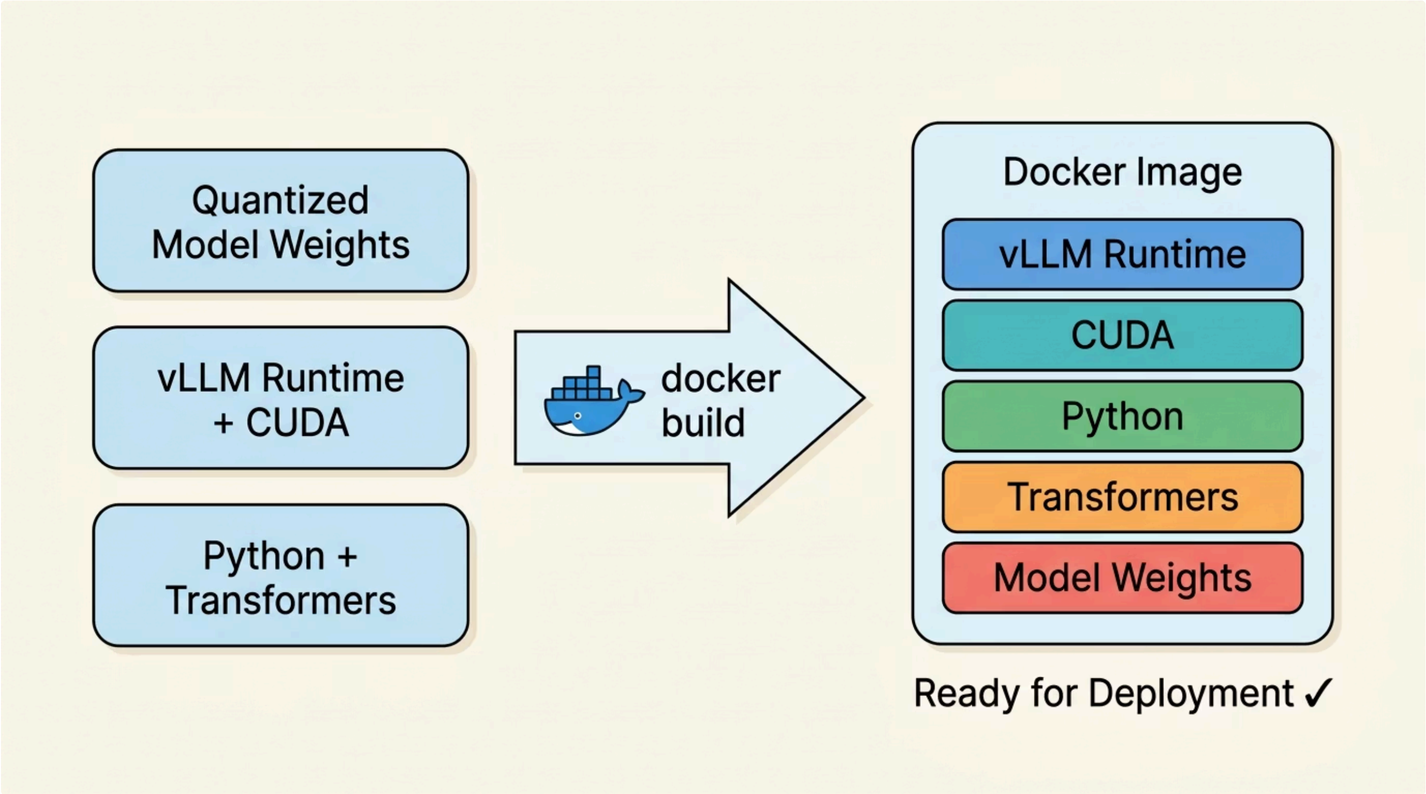


🐳 Step 3

Package with vLLM

Wrap everything into one portable Docker image.

> vLLM + CUDA + model weights = one image. Deploy anywhere.

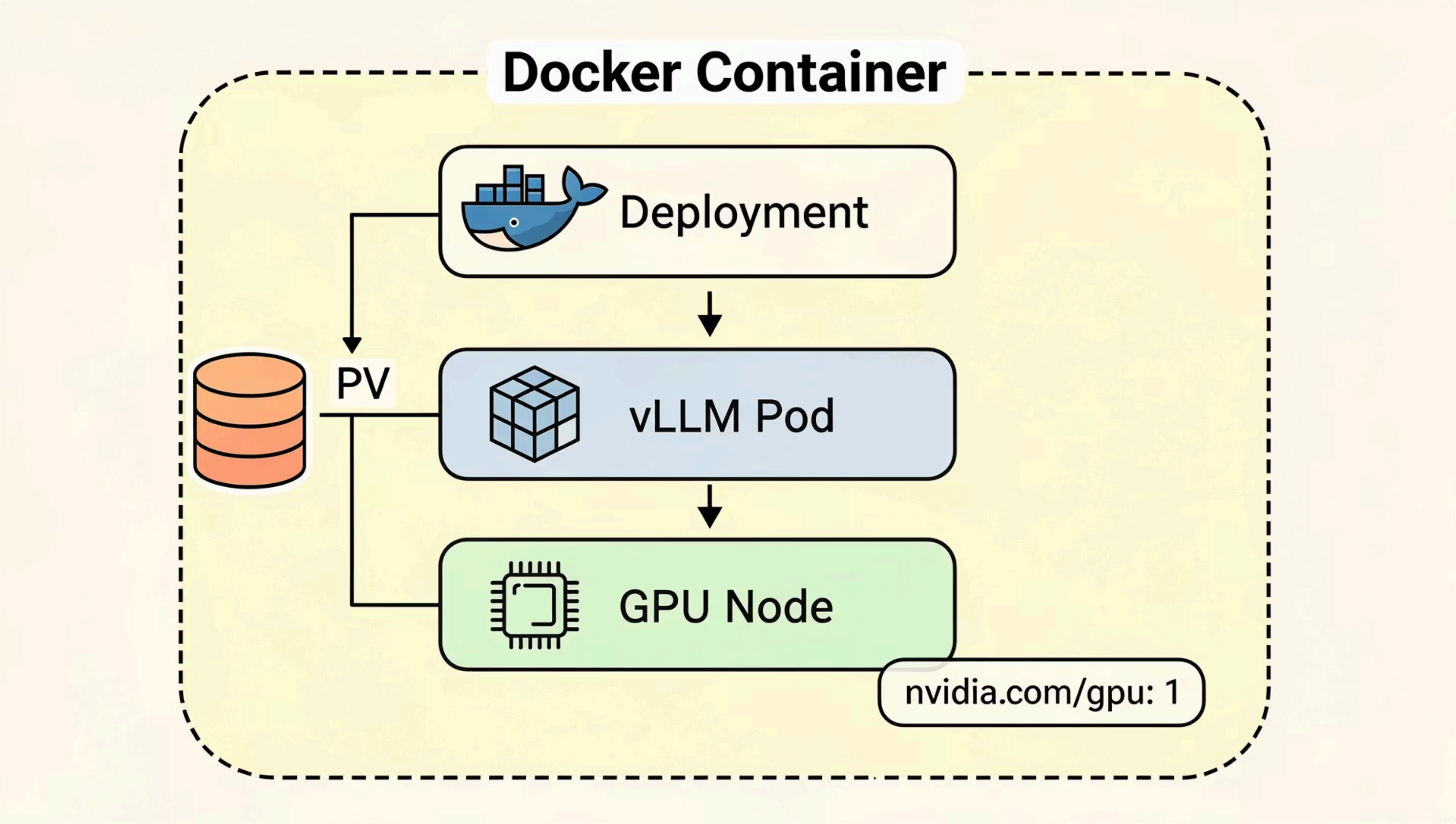


🐳 Step 4

Docker Container

Run the Docker container directly on a GPU-enabled machine.

> One container holds it all — Streamlit, vLLM, model weights, and CUDA runtime.

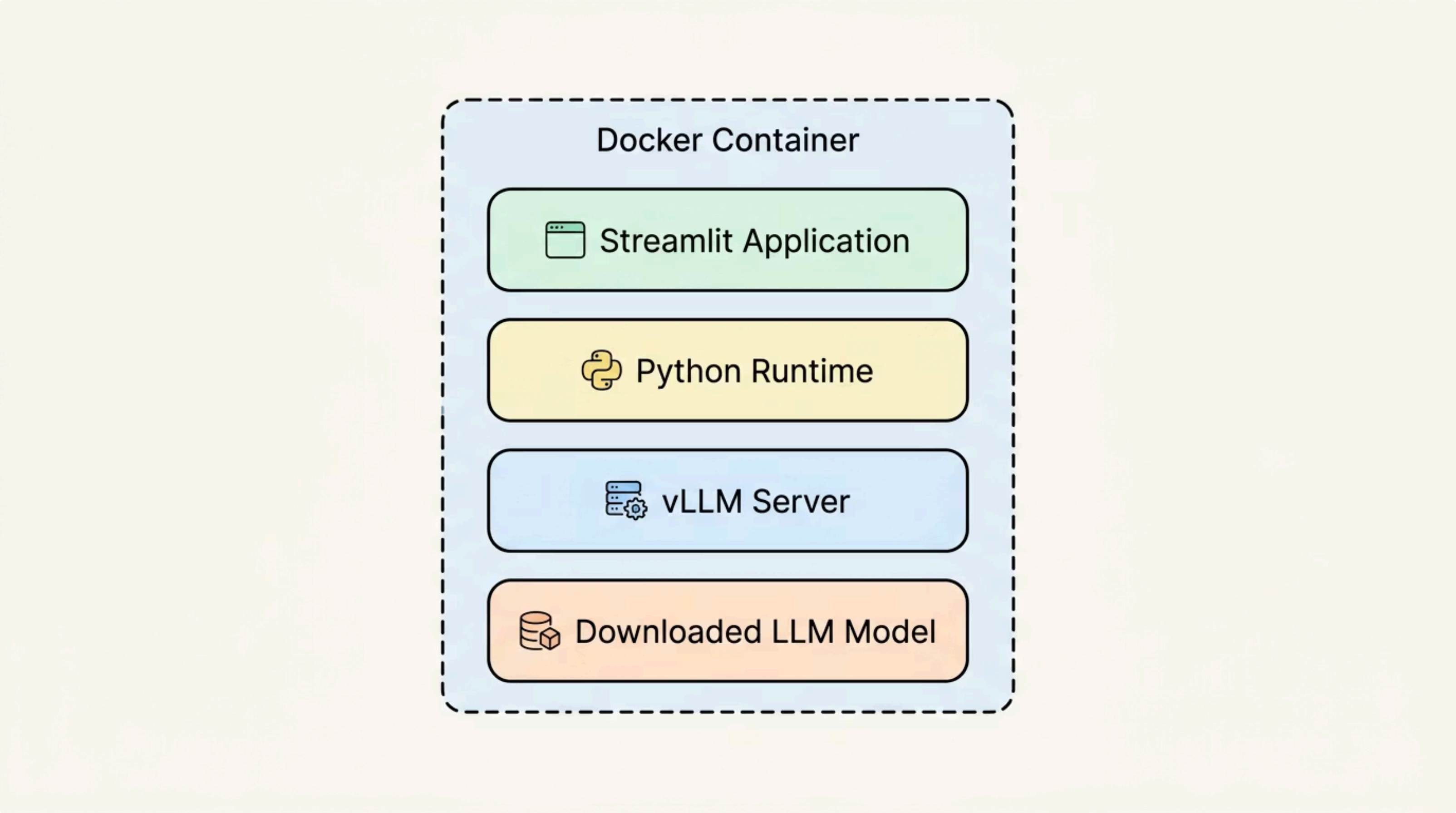


🐳 Step 5

Single Container Deployment

Package everything into one container — Streamlit, vLLM, and the model together.

> One container. One command. Streamlit + vLLM + model weights — all inside.



🖥️ Step 6

Deployment Target & User Access

Run the container on a GPU VM or cloud instance — users connect via browser.

> Browser → Streamlit → vLLM → GPU. One VM, one container, full inference pipeline.